

**Answer all questions.**

1. (a) The following diagram shows an outline of part of the Periodic Table.

**The letters shown are NOT the chemical symbols of the elements.**

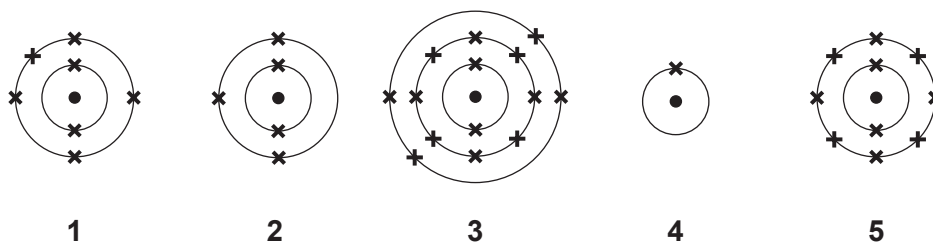
[illegible]

Choose **letters** from the diagram to complete the table below.

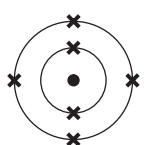
[5]

|                                                  | Letter(s)       |
|--------------------------------------------------|-----------------|
| Two elements in the same group                   | ..... and ..... |
| The element which has 12 protons in its nucleus  | .....           |
| The element with a full outer shell of electrons | .....           |
| The element in Group 2 and Period 2              | .....           |
| The element with the electronic structure 2,8,4  | .....           |

(b) Diagrams 1-5 show the electronic structure of five elements in the Periodic Table.



Give the **number** of the diagram which shows the electronic structure of the element which lies

(i) directly **below**  in the Periodic Table, ..... [1]

(ii) to the **left** of  in the Periodic Table. .... [1]

(c) Nitrogen has two stable isotopes – nitrogen-14 and nitrogen-15.

Describe how these isotopes are similar to one another and how they are different. [2]

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